

IIIT-PGEE



General Aptitude QUANTITATIVE APTITUDE

Lecture No.- 02



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Recap of Previous Lecture



Topics

✓
Calendars & Clocks



Topics to be Covered



Topic-1

Clocks ✓

Topic-2

Averages ✓

Topic-3

Percentage ✓

[MCQ]



#Q. At 2:15, the two hands of a clock make an angle of?

Assignment

A 22.5°

B 15°

C 16 $\frac{1}{3}$ °

D 7 $\frac{1}{2}$ °

$$2 \rightarrow 60^\circ$$
$$15 \times 5.5 \rightarrow 82.5^\circ$$

$$22.5^\circ$$

Angle ^u $\rightarrow$ Time?

April, 2001

0
Sunday

1st April, 2001

2000 $\rightarrow$ $\emptyset$

J - ~~3~~ A - X

F - 0

17 - ~~8~~

Wednesday?

4th $\cup$

11th $\cup$

18th $\cup$

25th $\cup$

Second Pattern:

0°

Angle $\rightarrow$ Time?



0° OR Coincide:

1 hr → 1 time

12 hrs → 11 times

24 hrs → 22 times



180° OR Opposite:

1 hr $\Rightarrow$ 1 time

12 hrs $\Rightarrow$ 11 times

24 hrs $\Rightarrow$ 22 times

Except
0° & 180°

1 hr $\Rightarrow$ 2 times



90° OR Right angle:

Any angle
except

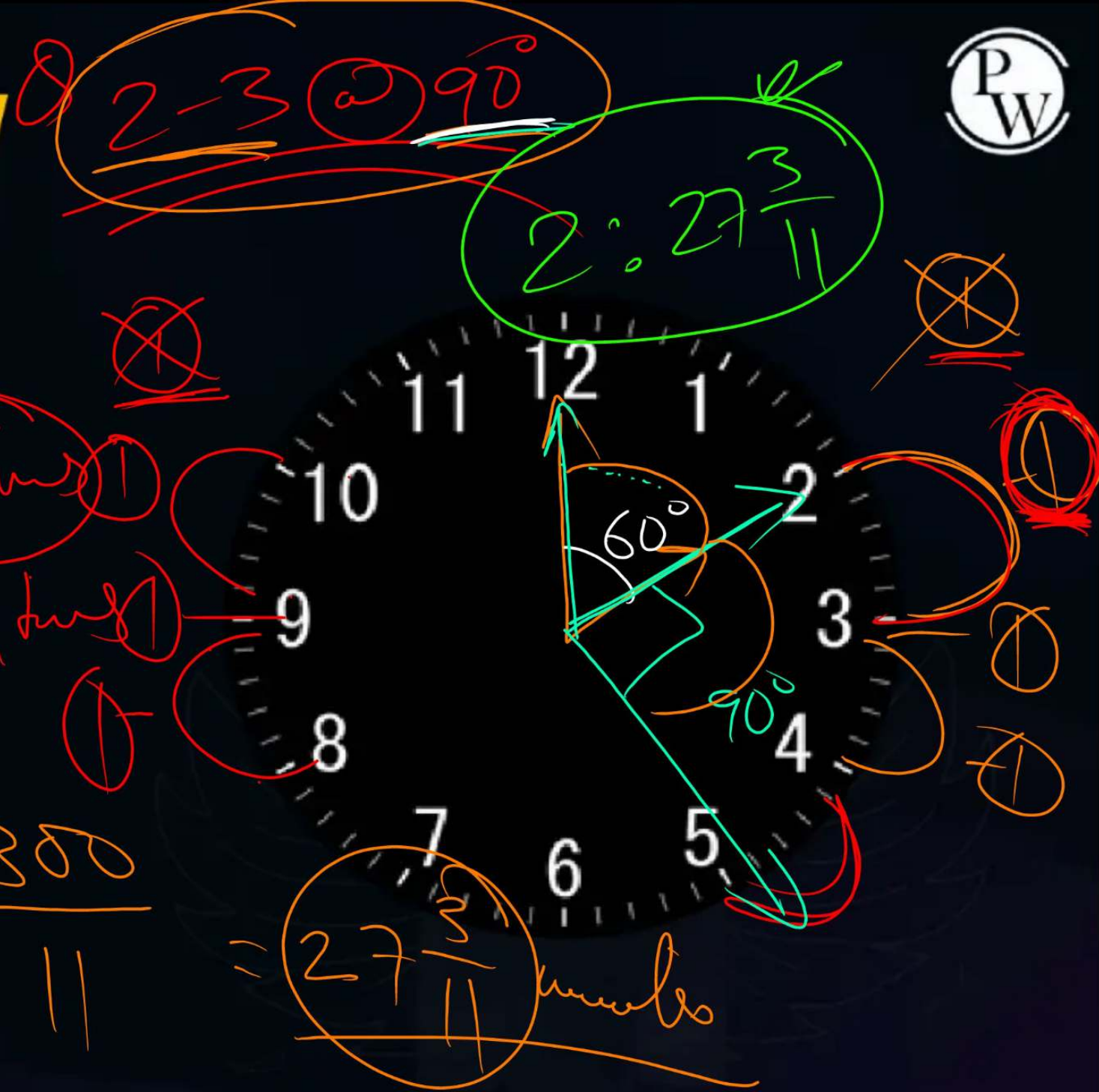
$$0^\circ < 180^\circ$$

$$12 \text{ hrs} = 22 \text{ times}$$

$$24 \text{ hrs} = 44 \text{ times}$$

$$\frac{150}{5 \times 5} = \frac{300}{11}$$

$$= 27 \frac{3}{11} \text{ minutes}$$



Questionnaire:



#Q. In between 5 0' clock and 6 0' clock at what time the hands of clock form 100°?

$$\frac{50}{5.5} = \frac{100}{11} = 9 \frac{1}{11} \text{ minutes}$$

$$\frac{250}{5.5} = \frac{500}{11} = 45 \frac{5}{11} \text{ minutes}$$



Questionnaire:



$$3:16\frac{4}{11}$$

#Q. Between 3:00 and 4:00, at what exact time do the hour and minute hands of a clock overlap?

$$\frac{90}{5-5} = \frac{180}{11}$$

$$= 16\frac{4}{11} \text{ minutes}$$



Questionnaire:



#Q. In between 6 O' clock and 7 O' clock at what time the hands of clock form

60°?

$$\frac{120}{5.5} = \frac{240}{11} = 21\frac{9}{11}$$

$$\frac{240}{5.5} = \frac{480}{11} = 43\frac{7}{11}$$

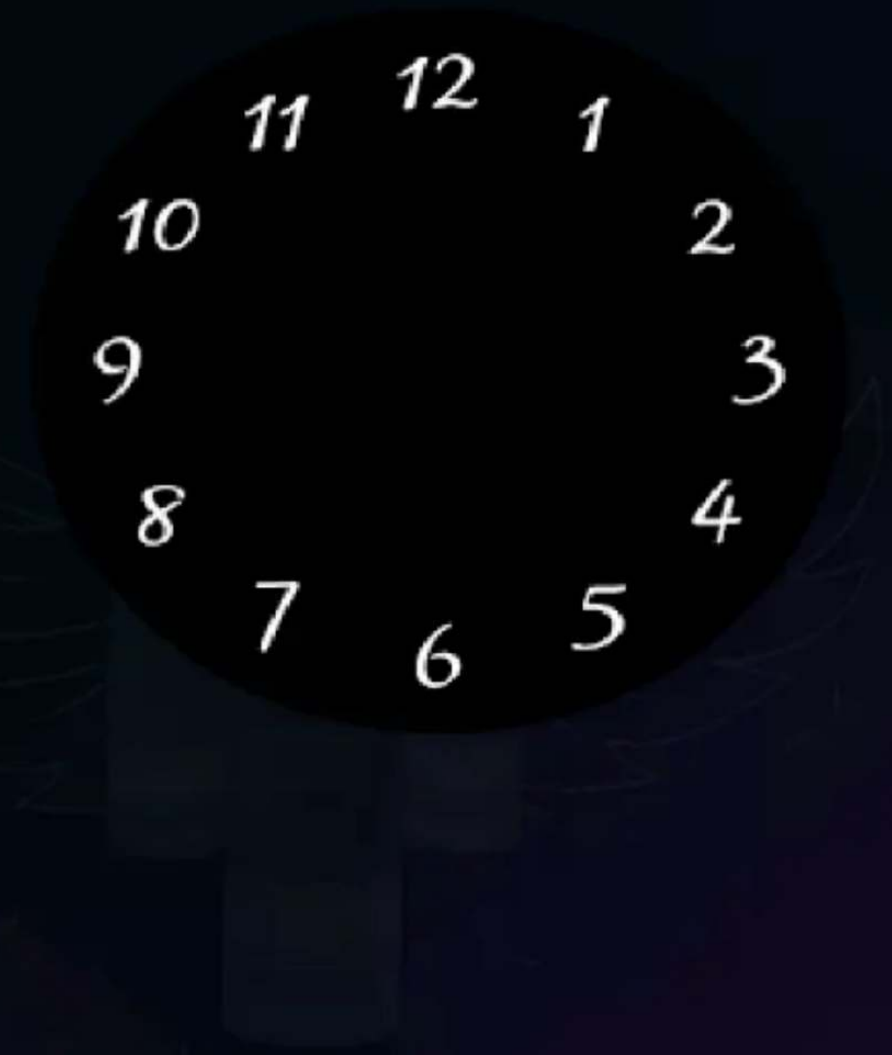


Questionnaire:

Assignment



#Q. At what time between 1 and 2 o'clock do the hands of a clock subtend an angle of 20° ?



Questionnaire:



7 hrs 30 minutes

7.5 hr

#Q. A worker noticed that the hour hand on the factory clock had moved by 225 degrees during her stay at the factory. For how long did she stay in the factory?

- A** 3.75 hours
- B** 4 hours 15 minutes
- C** 8.5 hours
- D** 7.5 hours

420 min

$$\frac{1^\circ}{2} \rightarrow 1 \text{ min}$$

$$1^\circ \rightarrow 2 \text{ min}$$

$$225^\circ \rightarrow 2 \times 225 = 450 \text{ min}$$

$$H \cdot H \rightarrow \frac{1^\circ}{2} \text{ minute}$$

Questionnaire:



6:45

360
 67.5
 292.5

#Q. It is quarter to seven in your watch. The angle between the hour hand and the minute hand is?

A 20°

B 75°

C 67.5°

D 292.5°

MSO

6 $\rightarrow 180^\circ$

$45 \times 5.5 \rightarrow 247.5^\circ$

67.5°



Questionnaire:



#Q. What is the difference, 11 hours after synchronisation, in the time shown by a standard watch and a watch whose hour and minute hands coincide every 64 minutes?

$$64 \text{ min} = 1 \frac{5}{11} \text{ hr}$$

$$65 \frac{5}{11} \text{ min}$$

A 11 minutes

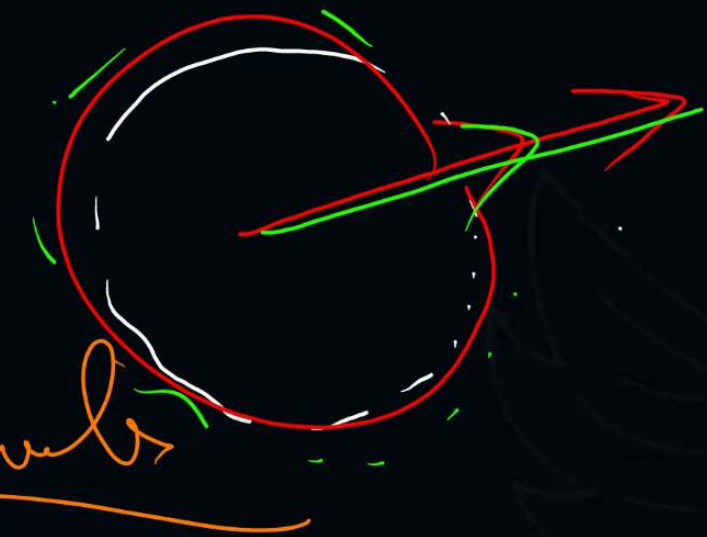
B 16 minutes

C 22 minutes

D 44 minutes

$$16 \times 11 = 176 \text{ min}$$

$$= 16 \text{ hours}$$



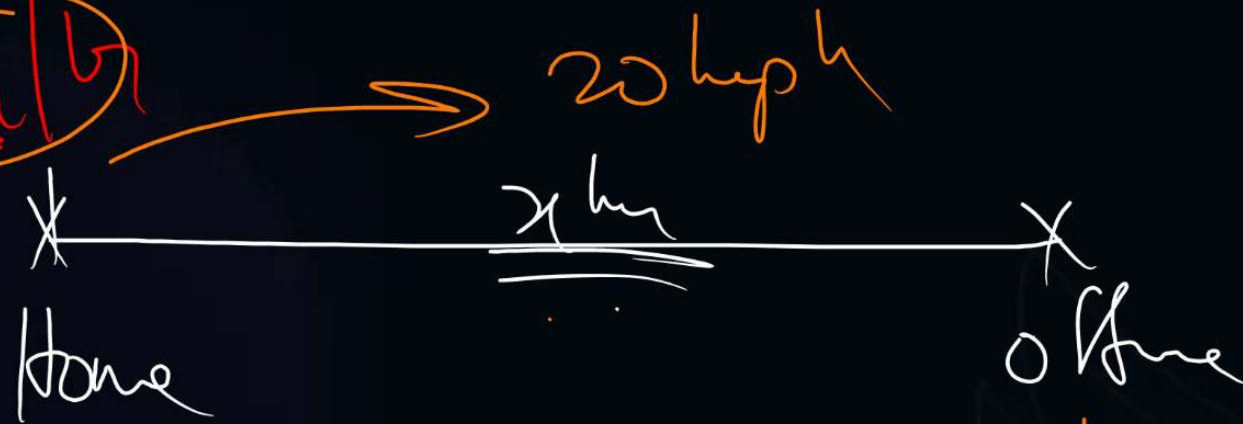
$$\frac{360}{5.5} = \frac{720}{11} = 65 \frac{5}{11}$$

AVERAGE



Sum / No
 Mean / Median / Mode
 25 hph
 Time = $\frac{D}{S}$

Average Speed
 24 hph



$$\frac{x}{20} + \frac{x}{30} = \frac{2x}{A.S.}$$

$$\frac{5x}{60} = \frac{2x}{A.S.}$$

$$A.S. = 12 \times 2 = 24$$

MEAN



$$\rightarrow \frac{\text{Sum}}{\text{No.}}$$

Equal distribution of given
values

MEDIAN

✓ Middle
✓ observation

~~15~~, ~~16~~, ~~18~~, 21, ~~09~~, ~~07~~, ~~18~~, ~~12~~

07, 09, 12, 15, 16, 18, 18, 21

15.5

$$\text{Mean} = \frac{\text{Sum of obs}}{\text{No. of obs}}$$

8

MODE

↓
formula observation

18

Average of Even & Odd

$$n=6$$

$$n=5$$

$$1, 3, 5 \rightarrow 3$$

$$2, 4, \boxed{6, 8}, 10, 12 \rightarrow 7$$

$$7$$

$$1, 3, 5, 7, 9 \rightarrow 5$$

$$2, 4, 6, 8, 10 \rightarrow 6$$

$$\frac{42}{6} = 7$$

$$\begin{aligned} \text{Even} &= n+1 \\ \text{Odd} &= n \end{aligned}$$

[MCQ]



#Q. What is the mean value of the first ten numbers divisible by 10?

$$\sum n = \frac{n(n+1)}{2}$$

A

10

B

100

C

50

D

55

$$\frac{50 + 60}{2}$$

10, 20, 30, 40, 50, 60, ..., 100

$$= 55 = \frac{10(1+2+3+\dots+10)}{10}$$

$$= \frac{10 \times 11}{2} = 55$$

[MCQ]



#Q. In a school, 60 students participated in a mathematics quiz. The average score of all the students was 72 marks. Later, it was found that 20 of those students had an average score of 84 marks. What was the average score of the remaining students?

A 64

B 69

C 66

D 68

Sum =

$$A = \frac{\text{Sum}}{\text{No.}}$$

$$A \times \text{No.} = \text{Sum}$$

$$\frac{2640}{40} = 66$$

$$72 - 6 = 66$$

$$72 \times 60 = 4320$$

$$84 \times 20 = 1680$$

$$2640$$

[MCQ]



#Q. Six years ago, the ages of three persons were in ratio 3:5:8. If the sum of their ages 8 years from now would be 122, what are their present ages (in years)?

- A 15, 25, 40
- B 15, 25, 30
- C 21, 26, 37
- D 21, 31, 46**

$$3x+14+5x+14+8x+14=122$$
$$15x+42=122$$
$$15x=80$$
$$x=5.33$$

$$14 \times 3 = 42$$

$$122 - 42 = 80$$

$$x = \frac{80}{16} = 5$$

$$21, 31, 46$$

[MCQ]

$M = 65$

Mode $\Rightarrow 65$



#Q. The following 15 observations are heights (in inches) of 15 persons

~~65~~, 61, 63, ~~65~~, 61, 61, 60, 60, ~~65~~, 85, ~~65~~, 86, 61, ~~65~~, 62

Which of the following is true?

Median = 63

60, 60, 61, 61, 61, 61, 62, (63), 65, 65
-8, -8, -4, -4, -4, -4, -3, -2

65, 65, 65, 85, 86
+20, +20
+16

Mean = $\frac{985}{15} = 65.\bar{6}$
+5

A

Mean < Median < Mode

B

Median < Mode < Mean

C

Mode < Median < Mean

D

Median < Mean < Mode

[MCQ]



#Q. In a field there are some cows and ducks. If the number of heads are 33 and number of legs are 84, then what would be the ratio of the numbers of cows to ducks?

$$\begin{aligned} C + D &= 33 \\ 4C + 2D &= 84 \end{aligned}$$

$$\begin{array}{r} 84 \\ 66 \checkmark \\ \hline 18 \end{array}$$

9 cows

$$\begin{array}{r} 24 \text{ ducks} \\ 9 \times 3 \\ \hline 24 \quad 8 \end{array}$$

- A** 3 : 8
- B** 5 : 6
- C** 2 : 5
- D** 3 : 5

[MCQ]

Assignment



#Q. The average of 5 consecutive integers starting with x is y . What is the average of 6 consecutive numbers starting with $(x+2)$?

- A** $y + 3$
- B** $\frac{2y + 9}{2}$
- C** $y + 2$
- D** $\frac{2y + 5}{2}$

[MCQ]

Assignment



#Q. A cricketer has certain average of runs for his 64 innings. In his 65th innings, he is bowled out for no score on his part. This brings down his average by 2 runs. His new average is?

- A** 130
- B** 128
- C** 70
- D** 68

PERCENTAGE

100

20% ↑

120% of original

(-)

92% of original 8% ↓

PERCENTAGE



$$120\% = \frac{120}{100} = 1.2$$

0.2144

Successive

$$\begin{aligned} & \xrightarrow{100} 120 \xrightarrow{10\% \text{ down}} 110.4 \xrightarrow{1\% \text{ down}} 121.44 \\ & \quad \quad \quad \downarrow \quad \quad \quad \downarrow \quad \quad \quad \downarrow \\ & \quad \quad \quad 20\% \uparrow \quad \quad \quad 8\% \downarrow \quad \quad \quad 10\% \uparrow \end{aligned}$$

$$1.2 \times 0.92 \times 1.1$$

$$= 1.2144$$

21.44% ↑

[MCQ]



#Q. Sujeet drinks 20% of the milk present in a bottle in one sip. How much milk will remain in the bottle after the first 3 sips, if it contained 1 litre milk to begin with?

A

512 ml

B

400 ml

C

625 ml

D

600 ml

$$0.8 \times 0.8 \times 0.8 \times 1000 \text{ ml}$$

$$= 0.512 \times 1000$$

$$= \underline{\underline{512 \text{ ml}}}$$

[MCQ]

Assignment



#Q. The population of a town is increasing at a uniform rate. If its population was 90,000 and 96,000 in 2022 and 2023 respectively, what would be its population in 2024?

- A** 1,02,000
- B** 1,02,400
- C** 1,02,720
- D** 1,02,960

[MCQ]

Assignment



#Q. 8% of the people eligible to vote are between 20 and 25 years of age. In an election 85% of those eligible to vote, who were between 20 and 25 actually voted. In that election number of person between 20 and 25, who actually voted, was what percentage of those eligible to vote?

- A** 4.2%
- B** 8%
- C** 6.4%
- D** 6.8%



2 mins Summary



Topic

✓
✓
Clocks, Averages, Percentage ?



THANK - YOU

